

Kaggle ML:

X: features (explanatory variables)
y: result to predict

sklearn:

- Define model: RandomForest, XGBoost...
- Fit: capture pattern from training data
- Predict: using new data
- Evaluate: with MAE for example

Functions: train_test_split, fit, predict, mean_absolute_error

Missing Data:

1. Drop cols with NAs when % in col is high
2. Imputation (mean, most frequent...)
 - ↳ SimpleImputer (fit, transform, set_output)

Categorical Data:

1. Drop categorical variables

2. Ordinal encoding:

↳ Ordinal Encoder

↳ good if high cardinal

Breakfast	Breakfast
Daily	3
Weekly	2
Never	1

3. One-hot encoding:

↳ One Hot Encoder

↳ bad if high cardinal

Red	Green	Yellow
1	0	0
1	0	0
0	1	0
0	0	1
0	0	0

Cross validation

- run modelling on different subsets of data: cross_val_score
- for SMALL datasets no extra computational burden
- for LARGE datasets less useful because less randomness
- good for validating model performance

Data Leakage

- Target leakage: predictors include data not available at prediction
- Train-Test contamination: running pre-processing before split for ex.
 - ↳ can check cross-val accuracy, should be $\approx 80\%$